

Cloudflare Zero-day: Accessing Any Host Globally

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Content

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Or, when .well-known went well past the WAF

There is a URL on almost every modern website that exists for machines, not people. It lives under `/.well-known/acme-challenge/` and, for a few seconds during certificate issuance, a robot visits it to check that you really control the domain. The visit is expected to be uneventful, a routine silent task. In this case, that quiet path got very loud!

This write-up tells the story of how traffic aimed at that certificate path could reach origins behind Cloudflare even when the rest of the application was blocked by customer rules, why that matters, how we proved it with restraint, and how the issue is now fixed. It is written for researchers who want details and for security leaders who need the big picture without a textbook.

A 60-Second Primer on ACME

ACME is the protocol that lets certificate authorities verify domain control. In the HTTP-01 method, the CA expects your site to serve a one-time token at `/.well-known/acme-challenge/{token}`. The CA fetches that token over plain HTTPS like any other client would; if the bytes match, the certificate is issued. The intention is strict and minimal: let the bot

read a small file at a very specific path, and nothing more. The ACME hallway is for a robot carrying a clipboard, not a crowd slipping past the guard.

The Observation that Didn't Fit

We were reviewing access posture on a set of applications where the WAF was configured to block the world and only allow specific sources. Everywhere else, the WAF did exactly what it said on the tin. But when we pointed a request at `/.well-known/acme-challenge/{token}`, the WAF stepped aside and the origin answered with its own voice. That single change in who was speaking - Cloudflare interstitial vs. origin framework - was the tell.

To make sure we weren't chasing a tenant-specific misconfiguration, we spun up our controlled demo hosts behind Cloudflare that default to blocked for normal traffic:

<https://cf-php.fearsoff.org/>, <https://cf-spring.fearsoff.org/>, and <https://cf-nextjs.fearsoff.org/>.

Under ordinary paths, you'll meet the block page. Aim the same hosts at an ACME challenge path with any true token and you'll get an origin-generated response, most often a framework 404. The difference is visible even without headers: one page is Cloudflare's, the other is unmistakably the app.

Signals We Trusted

On the demo hosts, we used small, idempotent requests and appended a harmless suffix to the token (for example, adding `/æ` after the token) to show that no real ACME file was required for the behavior to appear. In every case, we received origin responses where we should have been meeting the WAF.

Below are representative screenshots from the demo environment. The first image shows a normal request ending at Cloudflare's block page. The second shows the Custom Rule we used to block cf- hostnames in the demo. The next three show each demo host returning an origin generated 404 when the request targets the ACME path.

Block page (normal request)

[image: Image]

Custom Rule - block cf-* hostnames

For the demo, we created a rule that blocks any hostname containing cf-. In production, many teams block the public internet and allow only corporate VPN egress. This rule simulates that posture for our demo.

[image: Image]

Origin 404 (Next.js)

[image: Image]

Origin 404 (Spring)

[image: Image]

Origin 404 (PHP)

[image: Image]

How We Obtained a Stable Challenge Token

For repeatable demonstrations, we wanted a challenge token that would not disappear mid-test. Cloudflare's SSL/TLS Custom Hostnames feature lets you manage hostnames and certificates for third parties that CNAME to your domain. We added a custom hostname named `cf-well-known.fearsoff.org` and explicitly selected HTTP validation. The screenshots below show the addition flow and the resulting Pending Validation state.

[image: Image][image: Image]

We intentionally did not create a DNS record for `cf-well-known.fearsoff.org`, so issuance remained pending indefinitely. In that pending state, Cloudflare surfaces the HTTP-01 URL that a validation bot would eventually request, for example:

```
http://cf-well-known.fearsoff.org/.well-known/acme-challenge/yMnW0cR2yv0yW-...Jm5QksreNRDUmqKfKPTk
```

We did not complete validation and we did not place any challenge file at that path. The goal was a deterministic, long-lived token shape to anchor our WAF behavior tests without racing a real CA. With the token in hand, we could exercise the `/.well-known/acme-challenge/{token}` route across all Cloudflare hosts globally.

Why This Matters in Practice

WAF controls are meant to be the front door. When a single maintenance path bypasses that door, the definition of inside moves. In practical terms, the trust boundary for `/.well-known/acme-challenge/...` slid from the WAF to the origin. Once the origin is addressable from the internet - even for one route - ordinary bugs acquire a network path and ordinary pages become reconnaissance.

Here is how that shift looked on our demos.

Spring / Tomcat

Under a normal posture, actuator endpoints sit behind the WAF and internal network controls. Reaching the application through the ACME path changed that boundary. Using a well known traversal quirk in some servlet stacks (`..;/`), the request could land on `/actuator/env` and return process environment and configuration. That data often includes sensitive values - database URLs, API tokens, cloud keys - and it materially raises the blast radius of any mistake in the origin.

[image: Image]

Next.js

Server side rendering frameworks routinely ship server derived values to the client in order to hydrate the page. That is fine when the WAF owns the front door. When the origin answers directly, the same page can expose operational details that were never intended to be reachable from the public internet.

[\[image: Image\]](#)

PHP routing

Many PHP apps route all requests through `index.php` and use a query parameter to select a view. When that pattern carries an LFI bug, public reachability turns it into a file read. Asking for `../../../../etc/hosts` is enough to demonstrate the impact. In our demo, even the 404 flow was routed through `index.php`, which is why it surfaced additional pages once the origin began to speak directly.

[\[image: Image\]](#)[\[image: Image\]](#)

These demonstrations are the consequence, not the cause. The root issue was the WAF decision on a special path. Once that door opened, anything fragile inside the origin was suddenly one request away.

Not Just 404s - Account WAF Rules Ignored

To prove this was not just a detour to a framework 404, we configured account level WAF rules to block requests carrying a sentinel header. At the root path, the request with `X-middleware-subrequest:` was blocked as intended. The very same request aimed at the ACME path was allowed and served by the application. In other words, account rules that should have stopped the request were not evaluated for that path.

Why does that distinction matter? Many real applications make decisions based on headers or pass header values into downstream code. When WAF rules that police headers are skipped, entire classes of issues regain a route to the origin: header driven SQL concatenation in legacy code, SSRF and host confusion via `X-Forwarded-Host` or `X-Original-URL`, cache key poisoning when caches vary on headers, method override tricks with `X-HTTP-Method-Override`, and debug toggles wired to custom headers. The obvious question follows - how many apps still trust headers more than they should, and how many rely on the WAF to stand between that trust and the internet?

What Likely Happened and the Fix

Our working hypothesis during investigation was that requests under `/.well-known/acme-challenge/` were being evaluated on a different code path - an implicit exception, intended to help certificate validation, that executed before customer blocking controls. That would explain the consistent origin responses under the ACME path and the block pages everywhere else.

On October 27, 2025, Cloudflare deployed a fix. We re-tested the same patterns and observed the expected behavior: the WAF applied customer rules uniformly, including on `/.well-known/acme-challenge/*`. The boring path became boring again.

AI and the New Attack Surface

Vulnerabilities like this WAF bypass take on added urgency with evolving AI-driven attacks. Automated tools powered by machine learning can rapidly enumerate and exploit exposed paths like `/.well-known/acme-challenge/`, probing for framework-specific weaknesses or misconfigurations at scale.

For instance, an AI model trained to identify servlet traversal quirks or PHP routing bugs could chain this bypass with targeted payloads, turning a narrow maintenance path into a broad attack vector. Conversely, AI-driven security tools can help defenders by simulating these attack scenarios, as seen in our collaboration with the [Crypto\[.\]com Security Team](#), who used AI analytics to validate the issue. As origins become directly addressable, the race between AI attackers and defenders intensifies, making robust WAF controls more critical than ever.

Timeline

- October 9, 2025 - Submitted via HackerOne.
- October 13, 2025 - Vendor validation started.
- October 14, 2025 - Triaged by HackerOne.
- October 27, 2025 - Final fix deployed; re-test confirmed fixed.

Thanks

We wish to express our deep gratitude to [Jason Lau, CISO](#) and the **Crypto[.]com Security Team, who we approached first to help independently verify this zero-day vulnerability. Their technical expertise, AI security capability, speed, and responsiveness enabled close collaboration with [Matthew Prince, CEO](#) and the Cloudflare team** to expedite the development and testing of the patch. Thanks to our collective efforts, organizations worldwide are safer today.

Closing

The most dangerous bugs often start as routine details. A certificate robot's hallway should never become a side door. We appreciate the quick turn from investigation to fix and the collaboration throughout the process.

You can read about this bug on the [Cloudflare blog](#).

If you like the reading, leave a comment on [X](#).

<https://fearsoff.org>

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